Wine Quality: Predicting Vinho Verde Wine Quality Using Elastic Net and Random Forest

1. Introduction

Quality certification matters in the wine industry because it affects pricing, consumer trust, and marketability. The standard approach relies on sensory evaluation by trained experts, who rate attributes like aroma, flavour balance, and mouthfeel. But sensory evaluation is subjective and hard to reproduce. Cortez et al. (2009) noted that the relationship between physicochemical composition and sensory preference is far from well understood. This makes a strong case for computational models that use objective physicochemical measurements to predict perceived quality, which could supplement expert judgement and improve production consistency.

This study uses the Wine Quality dataset from the UCI Machine Learning Repository, originally collected by Cortez et al. (2009) from the Vinho Verde denomination in the Minho region of northern Portugal. The dataset comprises 1,599 red wines and 4,898 white wines, each described by 11 physicochemical properties — including fixed acidity, volatile acidity, residual sugar, chlorides, sulphates, alcohol content, and density. The response variable is a quality score (integer, range 3–9), obtained as the median rating from at least three blind tastings by certified sommeliers. Following the approach of Cortez et al. (2009), quality is treated as a continuous response for regression rather than a categorical label for classification, thereby preserving the ordinal distance between quality levels.

Two supervised learning methods are compared. Elastic Net is a regularised linear regression that combines L1 (Lasso) and L2 (Ridge) penalties, allowing variable selection and multicollinearity handling simultaneously. This improves on the unregularised multiple regression used by Cortez et al. (2009). Random Forest is a nonlinear ensemble of decision trees trained on randomised feature subsets; Dewi and Chen (2019) showed it performs well on wine quality data. Comparing a linear with a nonlinear approach tests whether the physicochemical-quality relationship is better captured by additive effects or by more complex interactions.

2. Data Cleaning and Exploratory Data Analysis

2.1 Data Cleaning

The red and white wine datasets were loaded separately from semicolon-delimited CSV files, augmented with a binary type indicator variable (red/white), and merged into a single dataset of 6,497 observations across 13 variables: 11 continuous numeric predictors, 1 two-level categorical predictor (type), and 1 integer-valued response (quality). Column names were standardised by replacing periods with underscores for coding consistency.

Data quality was checked in three steps. Missing value checks found zero missing entries across all 13 variables. Then, 1,177 duplicate rows were identified and removed, leaving 5,320 observations for analysis. Variable types were confirmed via str(): all physicochemical features are continuous numeric, type is a factor with two levels, and quality is an integer ranging from 3 to 9.

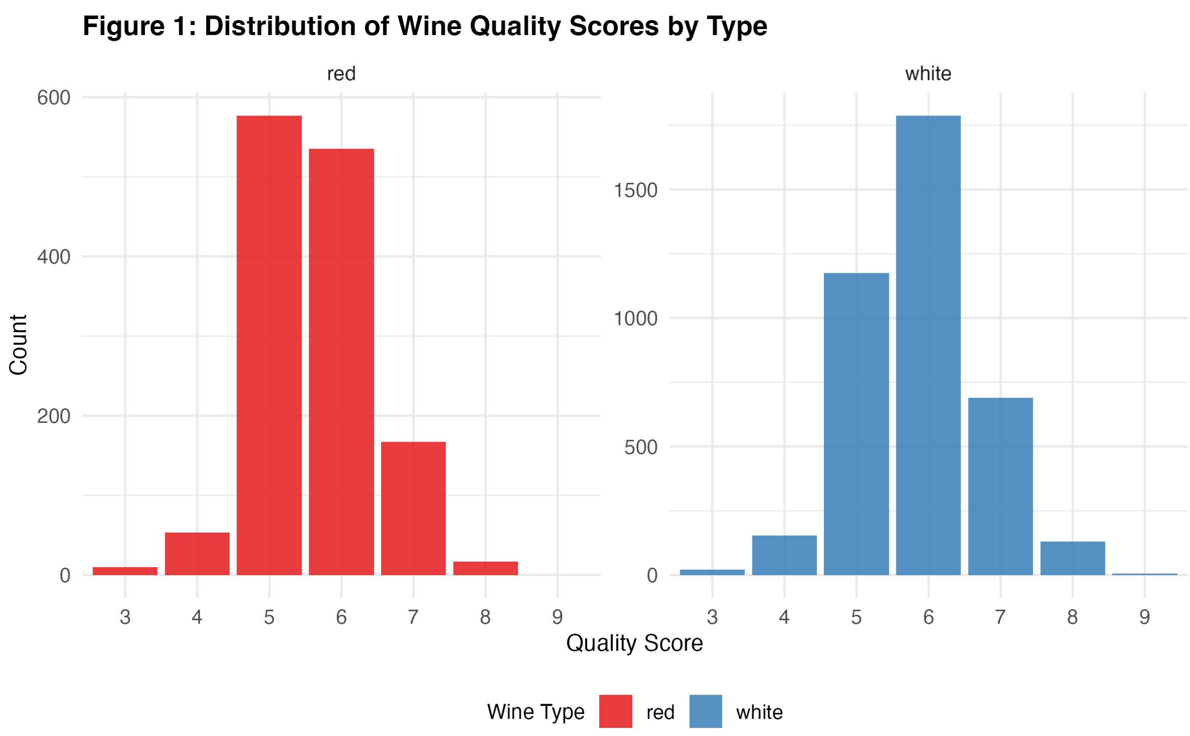

Figure 1. Distribution of wine quality scores by type.

The distribution of quality scores (Figure 1) reveals substantial class imbalance. Scores of 5 and 6 together account for 76.6% of all observations (1,752 and 2,323 samples respectively), while extreme ratings are rare — only 30 wines scored 3 and 5 wines scored 9. Red wines peak at quality 5, while white wines peak at 6. The mean quality is 5.80 with a median of 6. This concentration around the centre is consistent with the findings of Cortez et al. (2009) and reflects the reality that most commercially available wines are of moderate quality. Following Cortez et al. (2009), quality is treated as a continuous variable for regression, as the integer scores have a natural ordering and meaningful distance — a score of 7 is genuinely better than 6, not merely different.

2.2 Exploratory Data Analysis

Figure 2: Correlation Heatmap of Wine Features

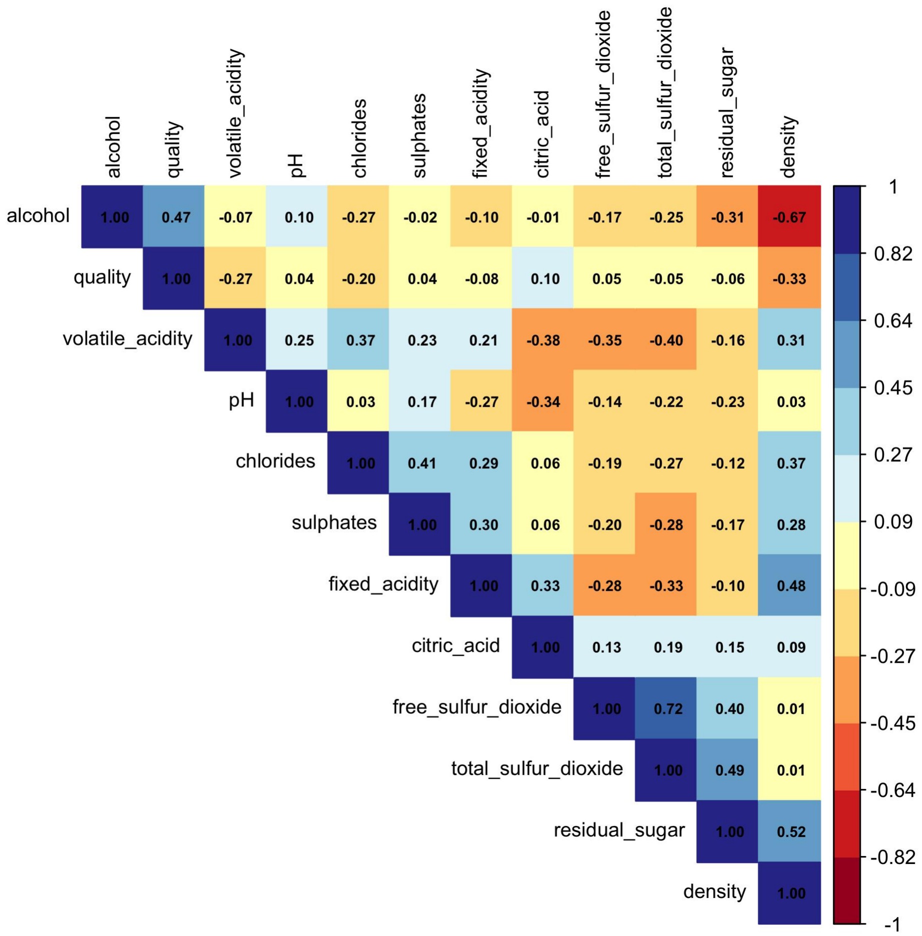

Figure 2. Pearson correlation heatmap of all numeric variables (hierarchical clustering order).

Figure 2 reveals that alcohol content has the strongest positive correlation with quality (r = 0.47), while density (r = −0.33) and volatile acidity (r = −0.27) are the most negatively correlated with quality. Notably, alcohol and density exhibit a strong negative correlation (r = −0.67), which is physically intuitive since ethanol is less dense than water — as alcohol content increases, overall density decreases. Furthermore, substantial multicollinearity exists among predictors, as evidenced by the correlations between density and residual sugar (r = 0.52) and between free and total sulphur dioxide (r = 0.72). This multicollinearity directly motivates the use of Elastic Net, whose L2 penalty stabilises coefficient estimates when predictors are correlated (Bhardwaj et al., 2022).

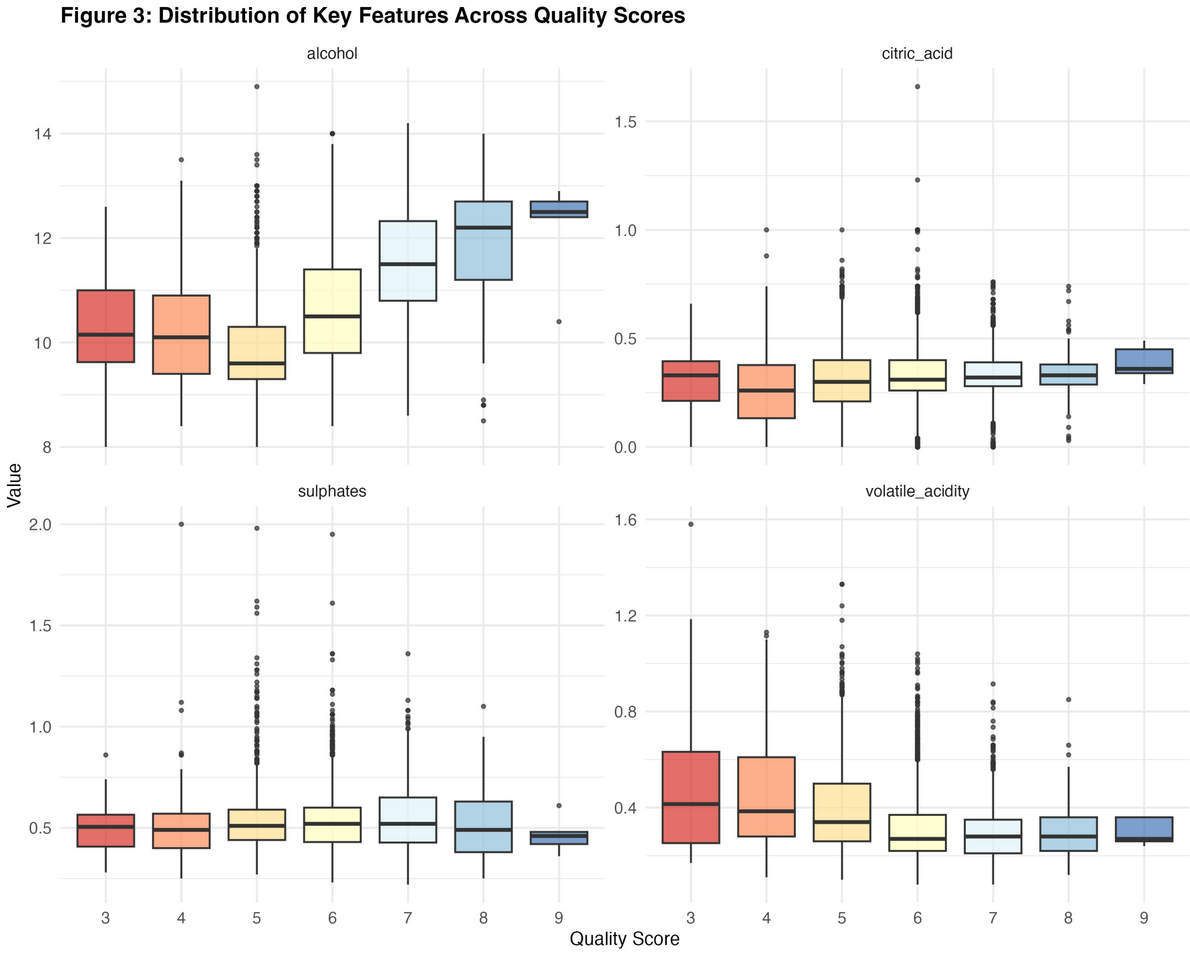

Figure 3. Boxplots of four key physicochemical features across quality scores.

Figure 3 shows clear trends. Most notably, once quality exceeds a score of 5, wine ratings increase significantly with rising alcohol content. Volatile acidity displays a clear negative relationship with quality: wines rated 3 have roughly twice the median volatile acidity of wines rated 8, consistent with the fact that excess acetic acid produces a vinegar-like off-flavour perceived negatively by tasters. Citric acid shows a slight upward trend as quality increases, suggesting that it contributes to a fresher, crisper palate and has a positive influence on perceived quality. Sulphates exhibit a notably tighter distribution among higher-rated wines, indicating that top-quality wines converge on more consistent sulphate levels. Therefore, the highest-quality wines are always in high alcohol content and low volatile acidity — whereas wines scoring around 5 contain numerous outliers, implying that high alcohol alone is insufficient for a high rating if other physicochemical attributes are not well balanced.

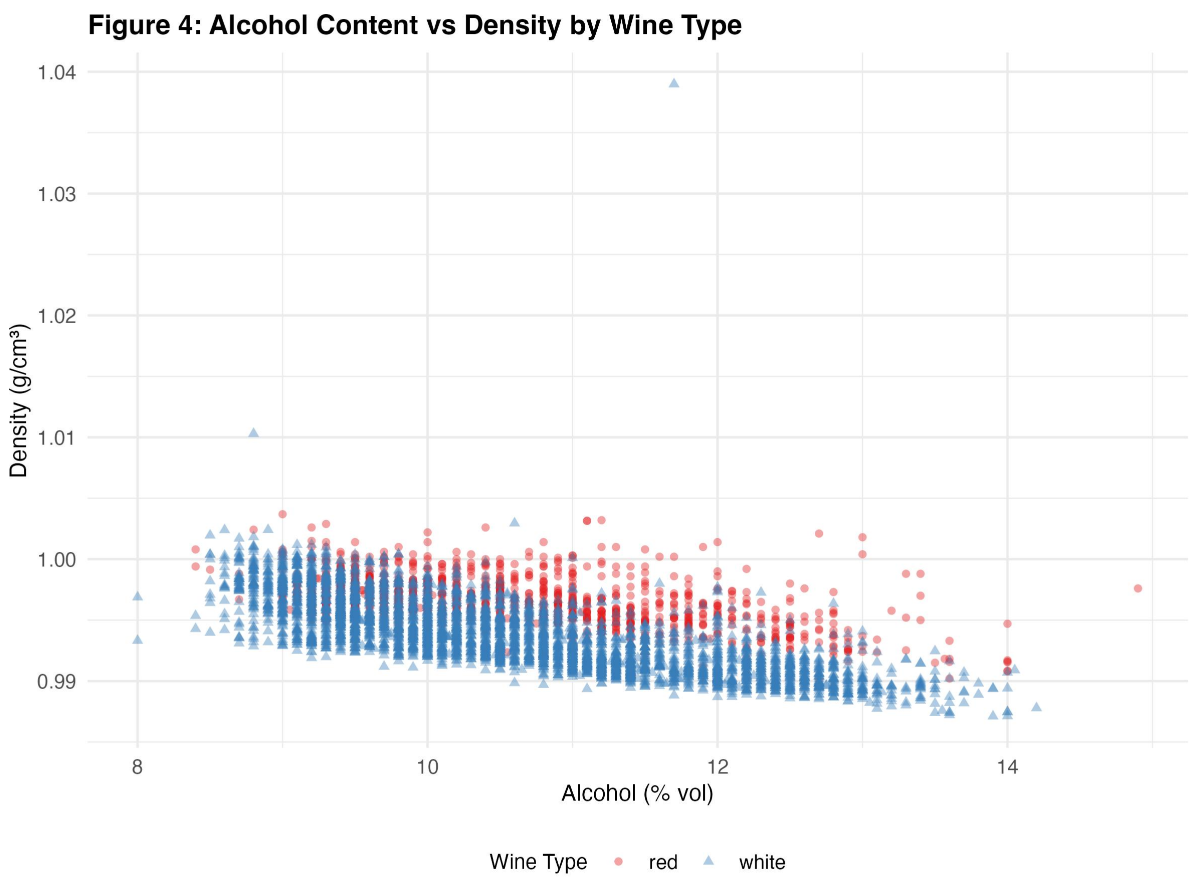

Figure 4. Alcohol content versus density by wine type. Red wines shown as circles (●); white wines as triangles (▲).

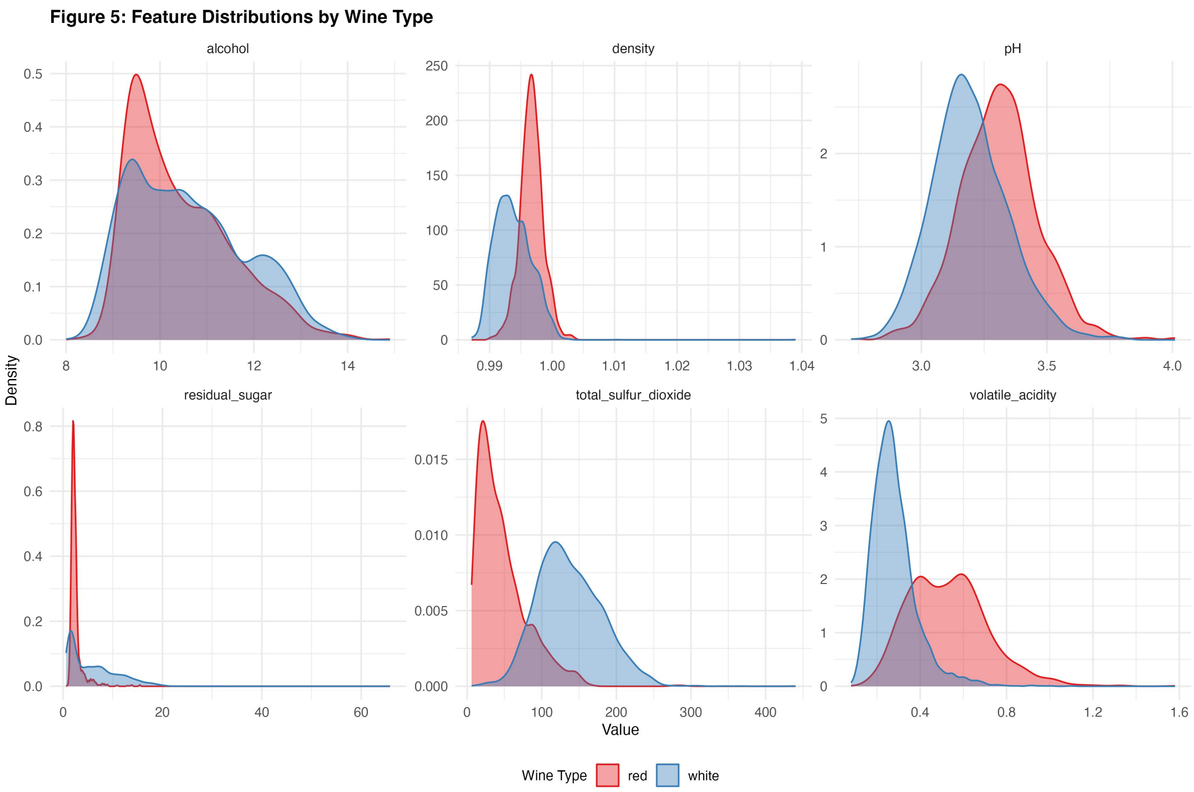

Figure 5. Kernel density plots comparing feature distributions between red and white wines.

Figure 4 shows a strong negative relationship between these two variables, with white wines generally clustering at lower density values for a given alcohol level. Figure 5 further reveal that red and white wines differ most dramatically in residual sugar (white wines exhibit a heavy right tail reflecting sweeter styles), total sulphur dioxide (substantially higher in white wines due to winemaking practices), and volatile acidity (higher in red wines). These systematic compositional differences justify the inclusion of type as a predictor rather than modelling the two wine types separately.

We can get three key insights guide the modelling strategy: (1) alcohol is the strongest single predictor of quality; (2) substantial multicollinearity among predictors necessitates regularisation for any linear model; and (3) the nonlinear patterns visible in Figure 3 suggest that a tree-based model may outperform a purely linear approach.

3. Modelling

3.0 Data Preparation

The dataset was split into training (4,258 samples, 80%) and test (1,062 samples, 20%) sets using stratified sampling via caret::createDataPartition, which preserves the quality score distribution in both sets. This is important given the class imbalance described in Section 2.1.

Elastic Net requires feature standardisation because its regularisation penalties are magnitude-sensitive. Without scaling, predictors on larger scales (e.g., total sulphur dioxide, range 6-440) would be penalised more heavily than those on smaller scales (e.g., chlorides, range 0.009-0.611). The standardisation parameters (mean and standard deviation) were computed from the training set only and then applied to both training and test sets. Computing them from the full dataset would constitute data leakage, where test-set information inadvertently influences training and leads to overly optimistic performance estimates (Bhardwaj et al., 2022). Random Forest does not need feature scaling because tree splits are based on rank-order comparisons, not absolute magnitudes.

To ensure a fair head-to-head comparison, identical 10-fold cross-validation folds were generated using caret::trainControl with a fixed random seed (42) and shared across both modelling procedures.

3.1 Elastic Net

Method overview. Elastic Net minimises a penalised residual sum of squares that combines L1 (Lasso) and L2 (Ridge) penalty terms. Two hyperparameters control the model. Alpha (α) sets the mixing ratio between L1 and L2: $\alpha = 0$ gives pure Ridge, $\alpha = 1$ gives pure Lasso.

Lambda (λ) controls how strong the regularisation is overall. A large λ shrinks coefficients aggressively toward zero; a small λ makes the model closer to ordinary least squares. Compared to the standard multiple regression in Cortez et al. (2009), Elastic Net has two advantages: it can eliminate irrelevant predictors entirely through L1 shrinkage, and it stabilises coefficient estimates for correlated predictors through L2 regularisation.

Figure 7: Elastic Net - Alpha Tuning via Cross-Validation

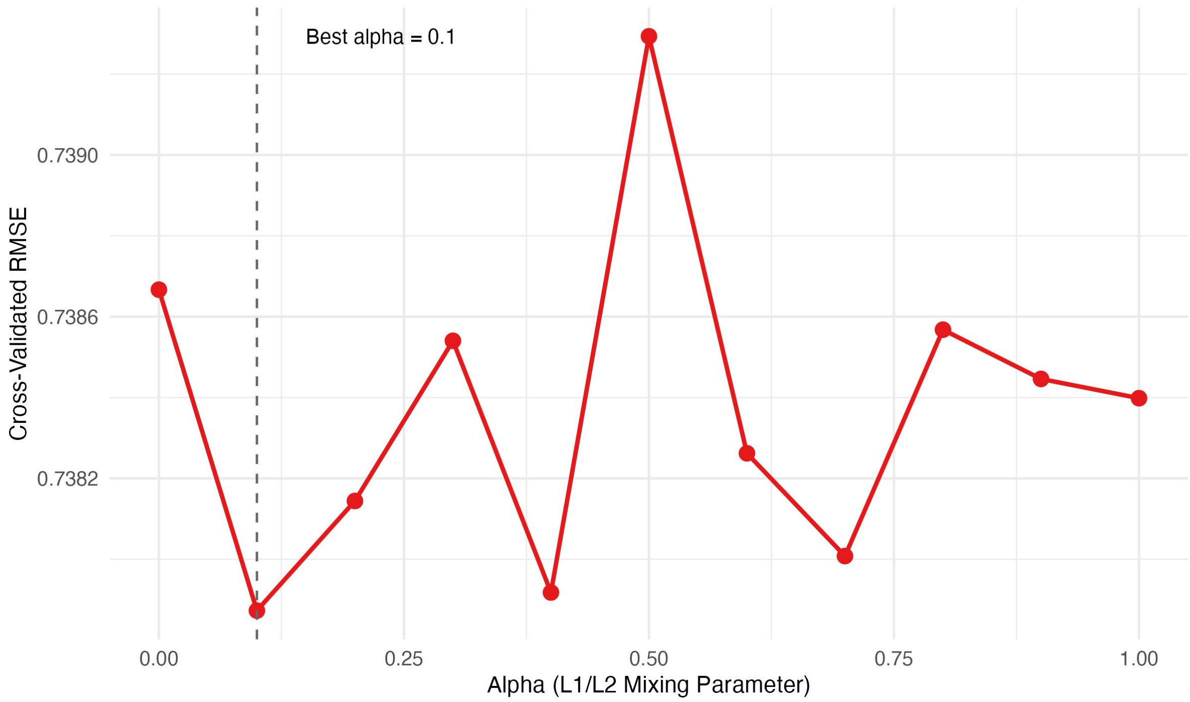

Figure 7. Elastic Net alpha tuning: cross-validated RMSE across 11 candidate alpha values. Dashed line indicates optimal α = 0.1.

Hyperparameter tuning. A two-stage grid search was used. First, 11 candidate α values were tested at equal intervals from 0 to 1 (step 0.1). For each α, cv.glmnet found the best λ via 10-fold cross-validation. Figure 7 shows that the lowest CV RMSE (0.7379) occurred at α = 0.1, meaning the model is mostly Ridge with only a small Lasso component. This makes sense given the multicollinearity in Figure 2: when predictors are correlated, Ridge works better than Lasso because it keeps all predictors with reduced coefficients instead of dropping one from a correlated group at random. The CV RMSE varied only slightly across α values (range: 0.7379–0.7393), so the linear signal is not very sensitive to the exact L1/L2 balance.

Figure 8: Elastic Net - Lambda Selection via CV

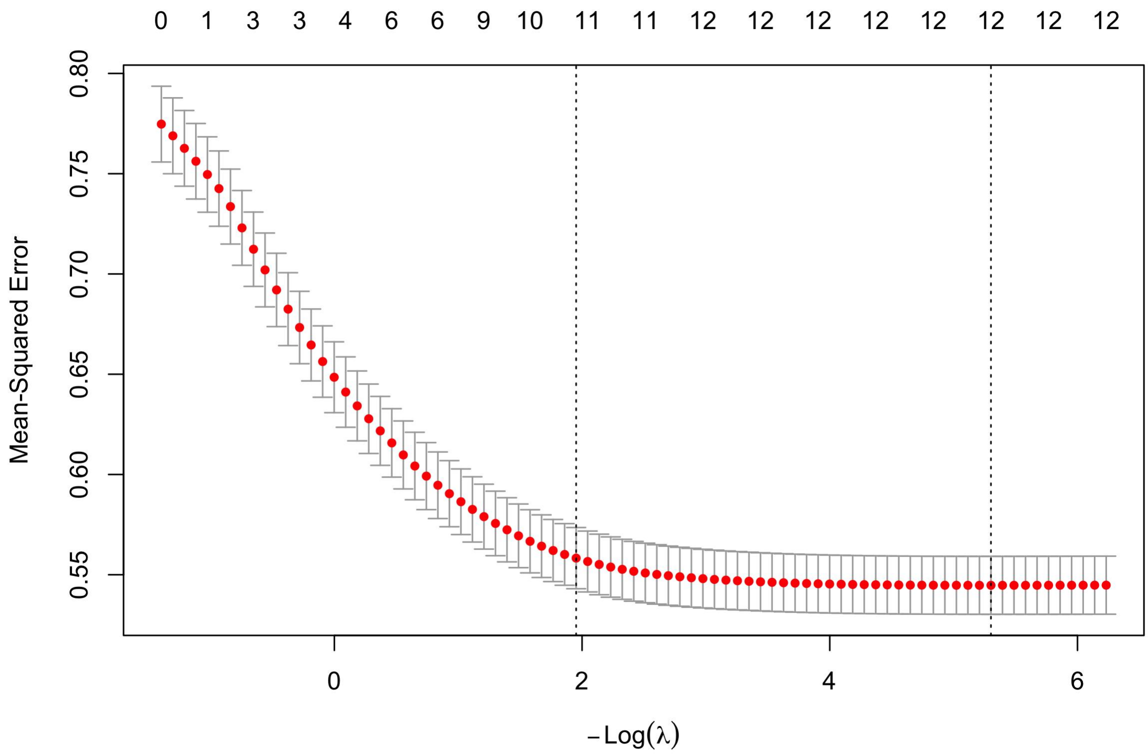

Figure 8. Cross-validation error as a function of log(λ) at α = 0.1. Left dashed line marks λ_min; right marks λ_1se.

At the optimal α = 0.1, the cross-validation error curve (Figure 8) yields a best λ of 0.006. As λ decreases (i.e., regularisation is relaxed), more coefficients become nonzero, and CV error drops sharply before plateauing around 12 active predictors.

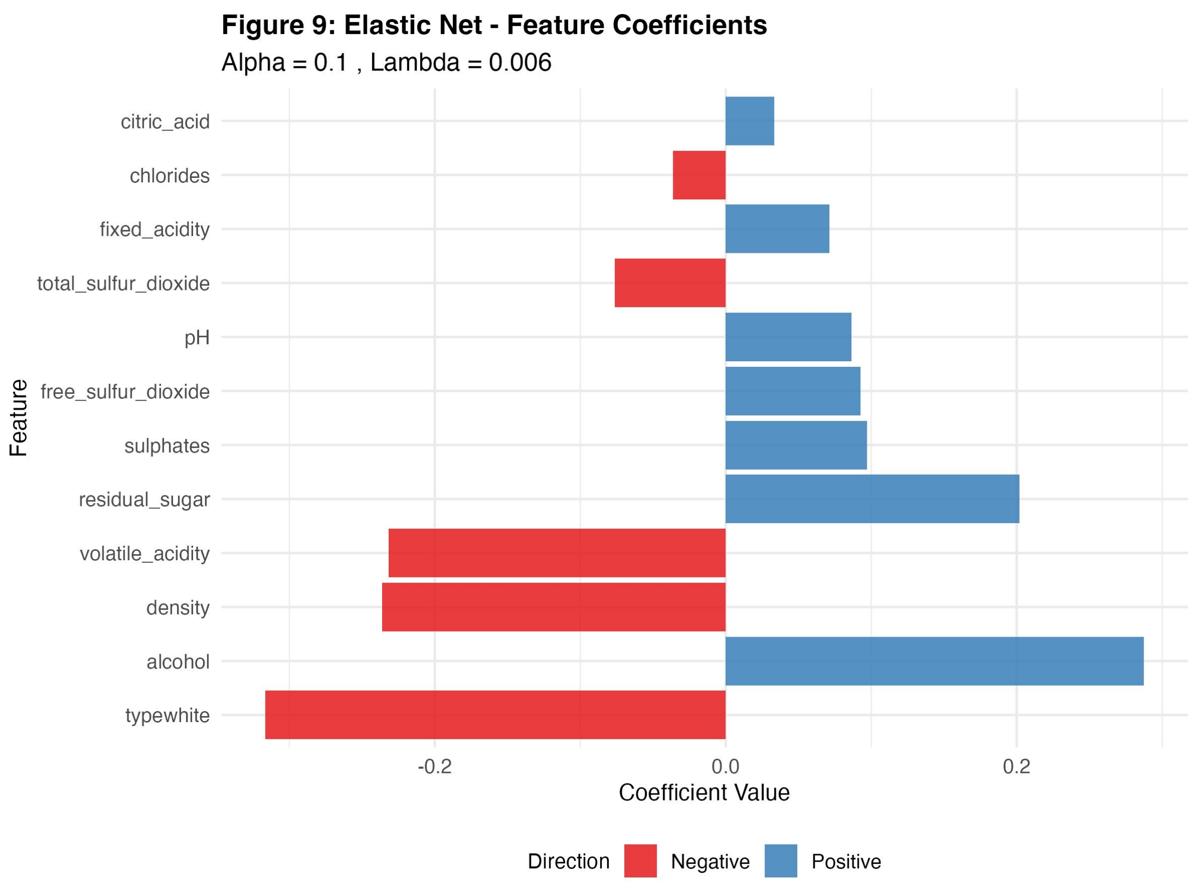

Figure 9. Standardised Elastic Net coefficients at α = 0.1, λ = 0.006. Blue bars denote positive coefficients; red bars denote negative coefficients.

Results. The standardised coefficient plot (Figure 9) shows what each predictor contributes. Alcohol has the largest positive coefficient. Other positive contributors are residual sugar, sulphates, free sulphur dioxide, pH, fixed acidity, and citric acid. The largest negative coefficients are wine type (white), density, and volatile acidity. No coefficients were shrunk to exactly zero. This is expected with $\alpha = 0.1$, since the L1 component is small and the model keeps all 12 predictors rather than eliminating any.

These directions match what is known about wine chemistry. Higher alcohol usually means riper grapes and fuller body, both of which tasters rate more highly. High volatile acidity means more acetic acid, a spoilage compound that gives wine a vinegar-like taste. Cortez et al. (2009) reached the same conclusion using sensitivity analysis on their SVM model, identifying alcohol and volatile acidity as the two most influential variables.

On the held-out test set, Elastic Net achieved: RMSE = 0.7120, MAE = 0.5447, $R^2 = 0.3423$.

3.2 Random Forest

Method overview. Random Forest builds many decision trees, each trained on a different bootstrap sample of the data. At each split, only a random subset of *m* features (out of *p* total) is considered, which decorrelates the trees and reduces the variance of the ensemble prediction. This two-layer randomisation (bootstrap samples plus random feature subsets) helps prevent overfitting and lets the model capture nonlinear relationships and interactions between features without having to specify them in advance. Dewi and Chen (2019) showed that Random Forest with feature selection outperformed SVM on this same dataset.

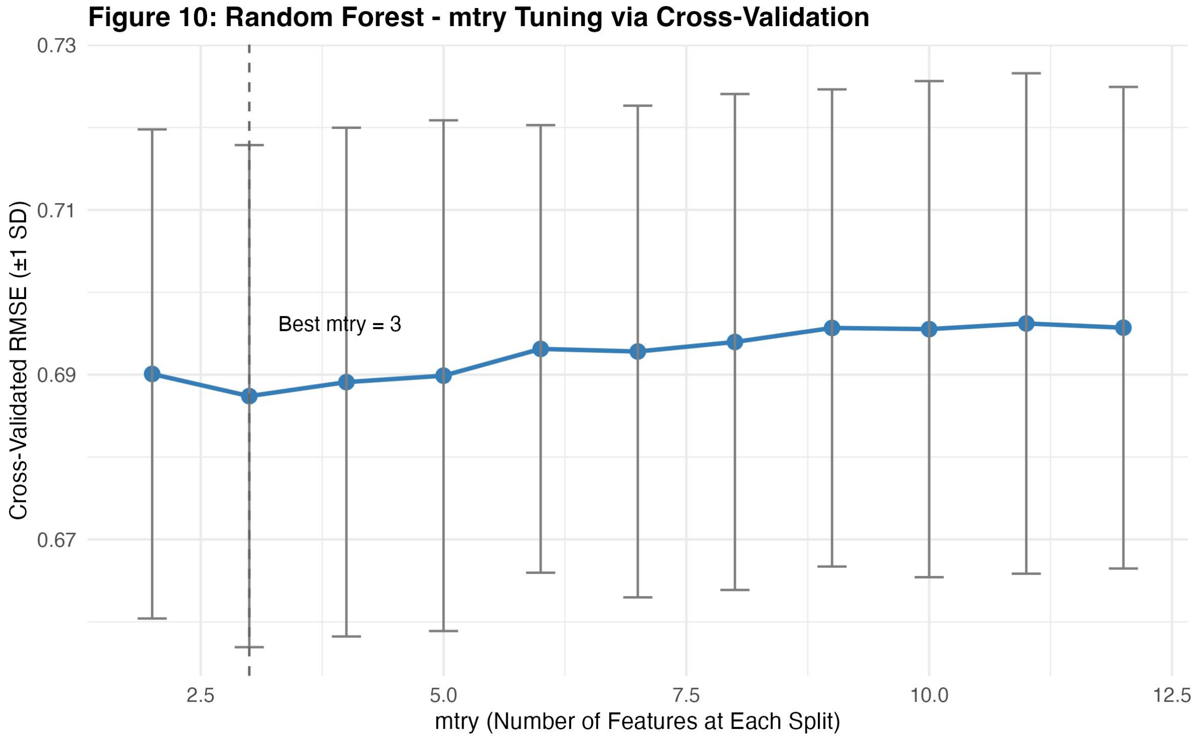

Figure 10. Random Forest mtry tuning: cross-validated RMSE (±1 SD) for mtry values 2 to 12. Dashed line indicates optimal mtry = 3.

Hyperparameter tuning. The main hyperparameter is mtry, the number of candidate features at each split. It was tuned over the range 2 to 12 using the same 10-fold CV folds. Figure 10 shows that CV RMSE is lowest at mtry = 3, slightly below the default $p/3 \approx 4$ for regression. The error bars overlap across mtry values, but lower values consistently do best, which suggests that using fewer features per split helps decorrelate the trees in this dataset.

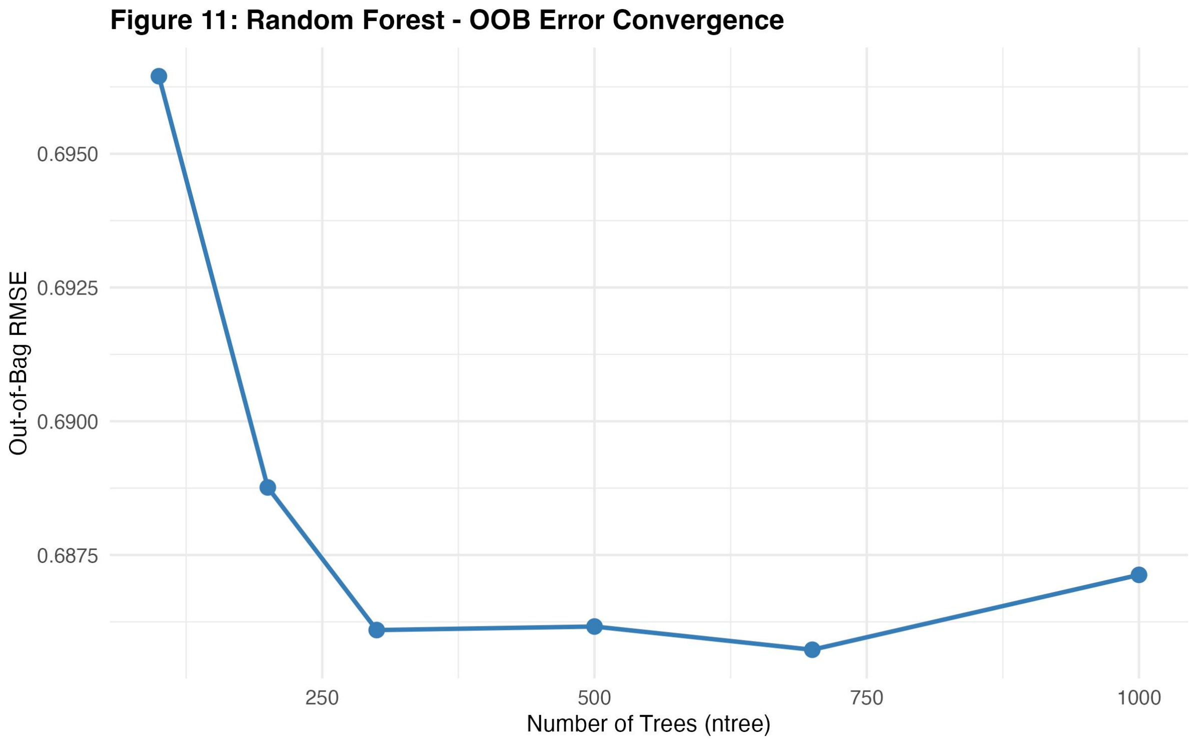

Figure 11. Out-of-bag RMSE as a function of the number of trees (ntree).

The effect of ntree was examined by training separate models with 100, 200, 300, 500, 700, and 1,000 trees. Figure 11 shows that OOB error decreases sharply from 0.696 at 100 trees to 0.686 at 300 trees, with negligible improvement beyond 500 trees. The final model uses mtry = 3 and ntree = 500, balancing predictive accuracy and computational cost.

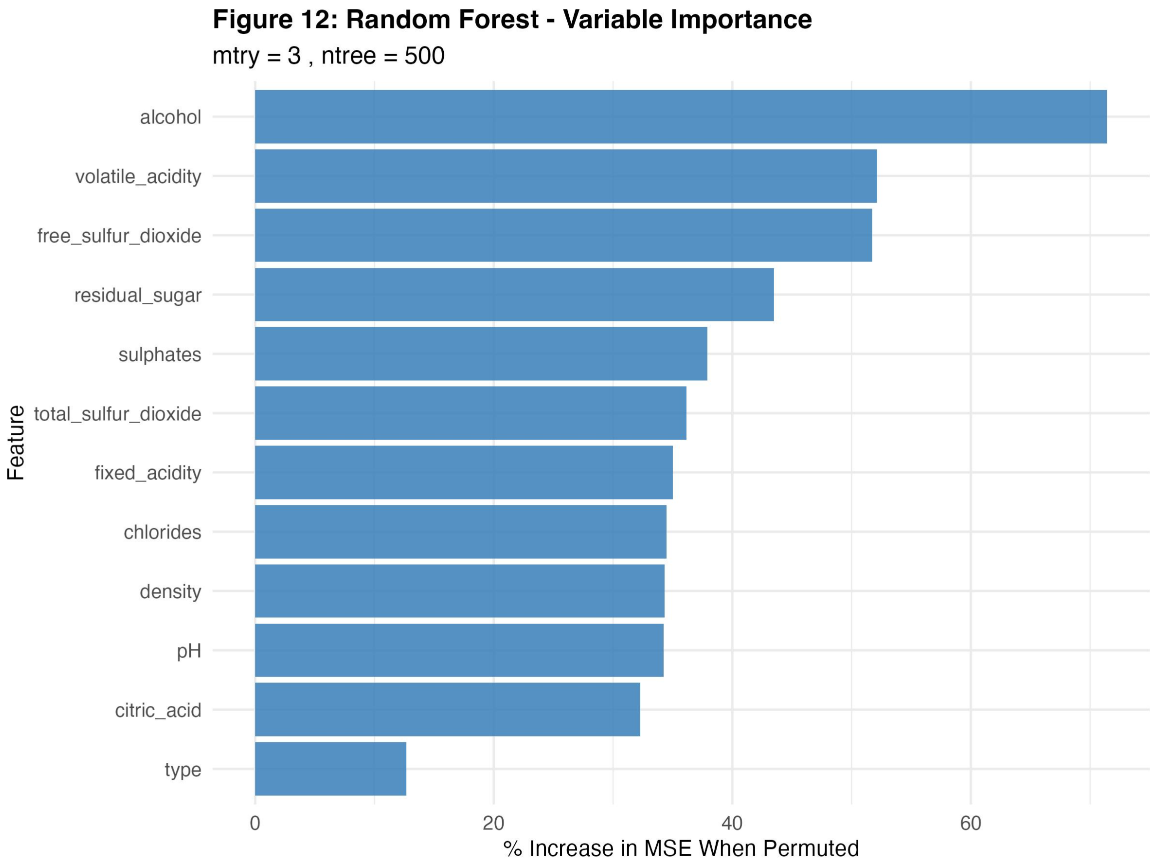

Figure 12. Random Forest variable importance, measured by %IncMSE (percentage increase in MSE when each feature is randomly permuted).

Results. Variable importance (Figure 12), measured by the percentage increase in MSE when each feature is randomly permuted (%IncMSE), puts alcohol well ahead of the other predictors (~73%), followed by volatile acidity (~53%), free sulphur dioxide (~52%), and residual sugar (~45%). This ranking matches Cortez et al. (2009), who found alcohol and volatile acidity to be the top predictors via SVM sensitivity analysis, and Dewi and Chen (2019), who reported alcohol as the top feature in their Random Forest model.

It is worth noting that the Random Forest importance ranking largely agrees with the Elastic Net coefficient magnitudes (Figure 9): both methods place alcohol and volatile acidity at the top. The fact that two fundamentally different models (one linear, one nonlinear) agree on the same top predictors suggests these physicochemical properties genuinely drive perceived quality, rather than being artefacts of any single algorithm. Bhardwaj et al. (2022) took a similar approach, using multiple models to cross-check variable importance in their study of New Zealand Pinot Noir.

On the held-out test set, Random Forest achieved: RMSE = 0.6715, MAE = 0.5082, R^2 = 0.4149.

4. Model Comparison

4.1 Methodology

Both models were trained on the same 10-fold cross-validation folds, so performance differences reflect model architecture rather than data partitioning. Three metrics are reported on the held-out test set. RMSE (root mean squared error) penalises large errors more because of the squaring step. MAE (mean absolute error) treats all error sizes equally and is less affected by outliers. R^2 (coefficient of determination) measures how much of the variance in quality scores the model explains.

4.2 Results

Metric	Elastic Net	Random Forest	Improvement
CV RMSE (mean)	0.7371	0.6874	−6.7%
Test RMSE	0.7120	0.6715	−5.7%
Test MAE	0.5447	0.5082	−6.7%
Test R^2	0.3423	0.4149	+21.2%

Table 1. Performance comparison between Elastic Net and Random Forest.

Figure 13: Cross-Validated RMSE Comparison

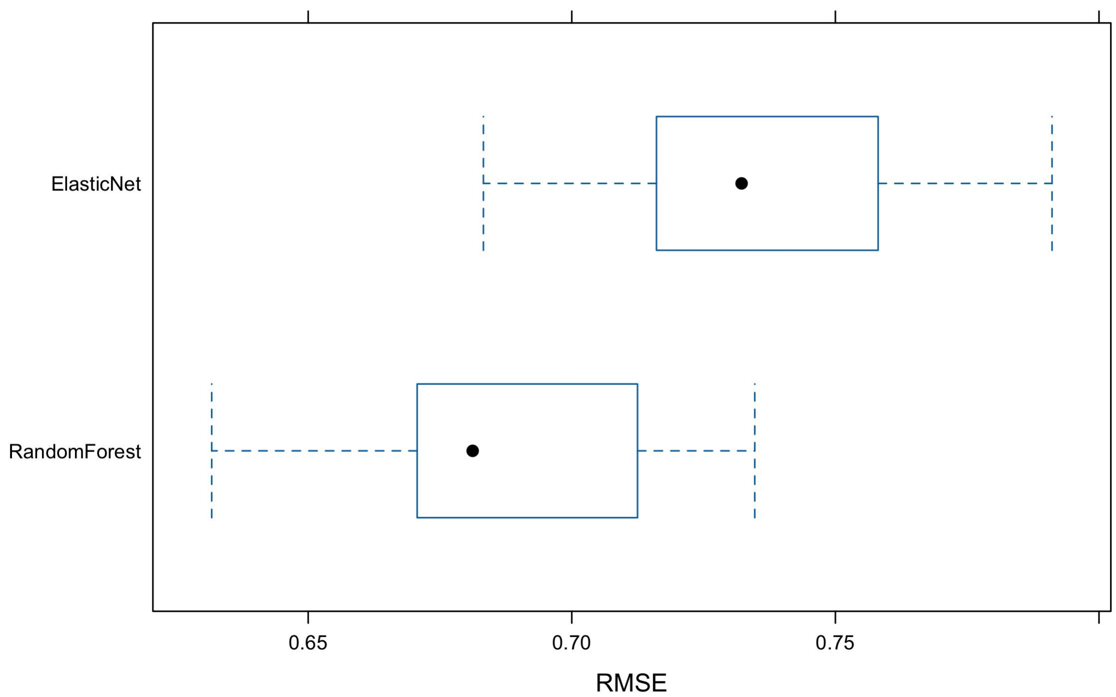

Figure 13. Boxplot comparison of cross-validated RMSE distributions.

Random Forest outperforms Elastic Net across all metrics, achieving a 5.7% reduction in test RMSE and explaining 7.3 percentage points more variance (R^2 of 0.4149 vs 0.3423). The CV boxplot comparison (Figure 13) confirms this pattern: Random Forest has a lower median RMSE with minimal overlap between the two distributions, indicating consistently superior performance across all 10 folds.

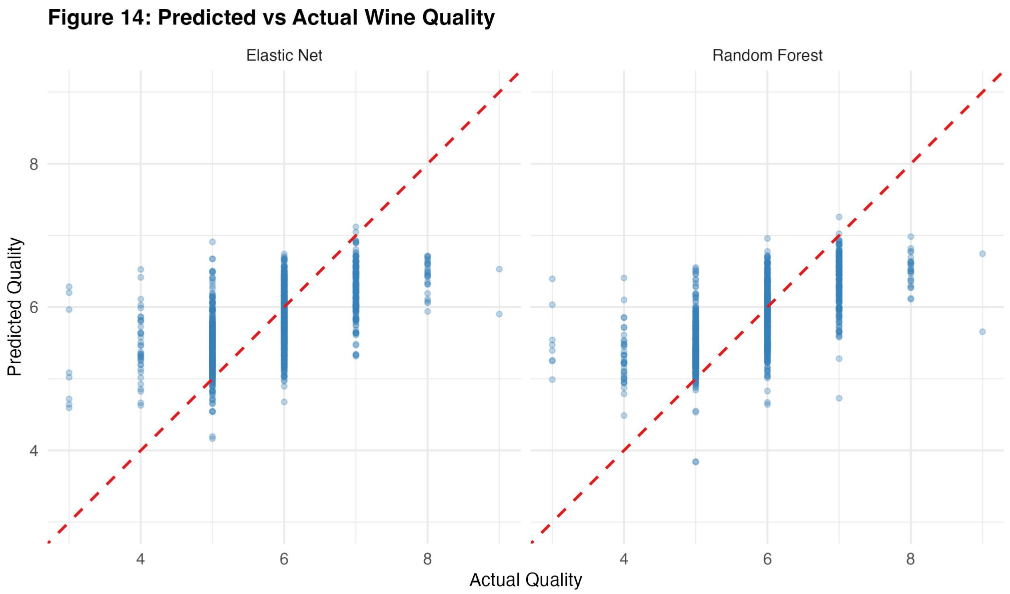

Figure 14. Predicted versus actual quality scores. The dashed red line represents perfect prediction (y = x).

The predicted-versus-actual plots (Figure 14) show that both models perform best for wines rated 5–7, where training data is abundant, but struggle with extreme scores. Elastic Net predictions are more compressed around the mean (approximately 5–7 range), a characteristic of regularised linear models that shrink predictions toward the centre. Random Forest predictions show greater spread and track the diagonal more closely for the central quality range. Neither model predicts values below approximately 4 or above 8, a direct consequence of the severe class imbalance at distribution extremes — a limitation also documented by Cortez et al. (2009).

Figure 15: Residual Analysis

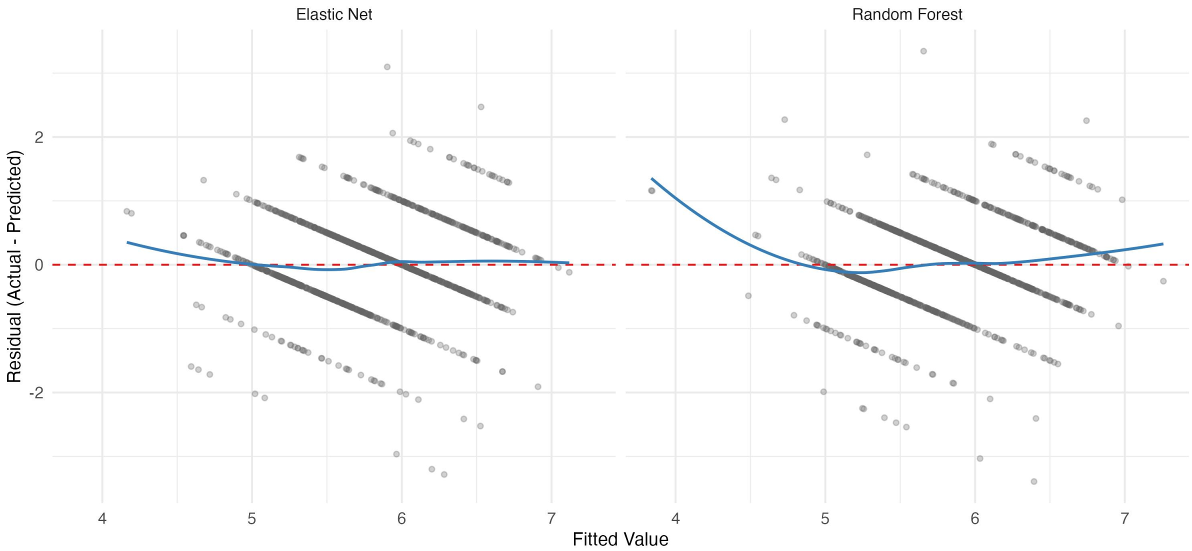

Figure 15. Residual plots for both models. The red dashed line marks zero residual; the blue curve is a LOESS smoother.

Residual analysis (Figure 15) shows that both models have systematic patterns, with the LOESS curves deviating from the zero line. Elastic Net residuals trend upward at higher fitted values, meaning it systematically underpredicts higher-quality wines. Random Forest residuals stay closer to zero across most of the fitted range, which indicates better calibration overall.

4.3 Discussion

Random Forest doing better is not surprising given the EDA findings. The nonlinear patterns in Figure 3 (for example, the diminishing effect of alcohol at higher quality levels) cannot be captured by Elastic Net's additive linear structure. Random Forest handles these nonlinearities naturally through recursive partitioning.

That said, Elastic Net has an advantage in interpretability. Its coefficient plot (Figure 9) tells you both the direction and size of each predictor's effect, which is useful if you want to advise winemakers on what to change. Random Forest's importance plot (Figure 12) only ranks predictors by importance without telling you the direction of the effect.

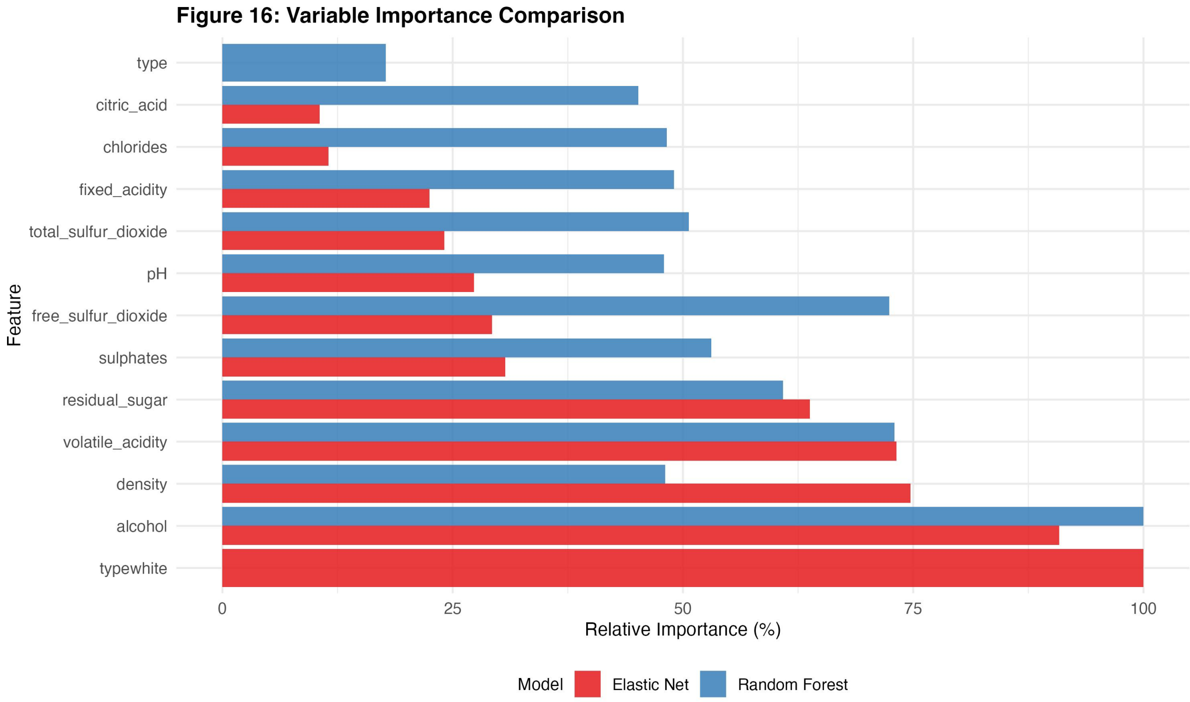

Figure 16. Normalised variable importance comparison. Elastic Net importance based on absolute standardised coefficients; Random Forest based on %IncMSE.

The variable importance comparison (Figure 16) shows that both methods agree on the top predictor (alcohol) but disagree on the type variable. Elastic Net gives it the highest importance, while Random Forest ranks it lowest. The reason is that Random Forest already captures red/white differences implicitly through the chemical features that vary between the two types (residual sugar, total sulphur dioxide; see Figure 5), so the explicit type indicator adds little. Elastic Net, being linear, needs the binary coefficient to account for these differences.

For context, Cortez et al. (2009) reported SVM MAD values of 0.45-0.46 for white wine alone, and Dewi and Chen (2019) reported SVM RMSE of 0.61 on the combined dataset. Our Random Forest RMSE of 0.6715 is in a comparable range, especially considering that analysing both wine types together introduces more variability.

5. Results and Conclusions

5.1 Key Findings

This study compared Elastic Net and Random Forest for predicting Vinho Verde wine quality from 11 physicochemical properties. Random Forest performed better on the test set (RMSE = 0.6715, R^2 = 0.4149) than Elastic Net (RMSE = 0.7120, R^2 = 0.3423), confirming that the relationship between chemical features and quality is nonlinear.

Both methods point to the same top predictors. Alcohol is the most important factor, followed by volatile acidity. Higher alcohol goes with higher quality ratings; higher volatile acidity goes with lower ratings. These results are in line with both the wine science literature and earlier computational work on this dataset (Cortez et al., 2009; Dewi & Chen, 2019). For winemakers, the implication is straightforward: achieving the right alcohol level through grape maturity management and keeping volatile acidity low through microbial control during fermentation are the two most effective levers for improving quality ratings.

The optimal $\alpha = 0.1$ for Elastic Net (mostly Ridge) means that keeping all features with reduced coefficients works better than aggressively dropping variables when the predictors are correlated. Elastic Net is less accurate than Random Forest, but its coefficients are directly interpretable, which matters in quality control settings where transparency is needed.

5.2 Limitations

There are several limitations to acknowledge. Quality scores, even though they are medians from multiple judges, remain subjective. The heavy class imbalance (76.6% of wines rated 5 or 6) limits both models' ability to predict extreme quality levels well. The dataset only covers the Vinho Verde region of Portugal, so the results may not apply to other grape varieties, climates, or winemaking traditions. And only 11 physicochemical features are available; a richer set of chemical measurements, like the 54-feature dataset that Bhardwaj et al. (2022) used for New Zealand Pinot Noir, might improve prediction accuracy.

5.3 Future Work

Two directions are worth exploring. Gradient boosting methods like XGBoost could be benchmarked against Random Forest, since boosting tends to do well on structured tabular data. Alternatively, the quality scores could be grouped into categories (e.g., Low: 3-4, Medium: 5-6, High: 7-9) and treated as a classification problem with oversampling techniques like SMOTE, as Bhardwaj et al. (2022) did. Another option is model stacking, where Elastic Net predictions are fed as an additional input to Random Forest, combining the linear model's global trend with the ensemble's ability to capture local nonlinearities.

References

Bhardwaj, P., Tiwari, P., Olejar, K., Parr, W., & Kulasiri, D. (2022). A machine learning application in wine quality prediction. *Machine Learning with Applications, 8*, 100261. https://doi.org/10.1016/j.mlwa.2022.100261

Cortez, P., Cerdeira, A., Almeida, F., Matos, T., & Reis, J. (2009). Modeling wine preferences by data mining from physicochemical properties. *Decision Support*

Systems, 47(4), 547–553. https://doi.org/10.1016/j.dss.2009.05.016

Dewi, C., & Chen, R.-C. (2019). Random forest and support vector machine on features selection for regression analysis. *International Journal of Innovative Computing, Information and Control, 15*(6), 2027–2037. https://doi.org/10.24507/ijicic.15.06.2027

Appendix

Declaration of Use of AI Tools in the Writing Process

AI tools (Claude, Anthropic) were used in the following capacities during the preparation of this report:

Language polishing: The AI tool provided suggestions for improving the readability, grammar, and clarity of the English text across all sections of the report. The author reviewed and modified all suggestions before incorporating them.

Affected pages: all pages. Tool: Claude (Anthropic). Purpose: code debugging assistance and language polishing.

Additional Figures

Figure 6: Pairwise Scatter Plot of Key Features